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Problem Set 3

I. PRINCIPAL PROTECTED NOTE

In this problem, we consider a so-called *principal protected note (or p.p.n.)*, the simplest capital guaranteed product offered by most banks to their retail clients.

The p.p.n. works as follows. A client is guaranteed to receive his money back at maturity plus a percentage ρ of the returns on an underlying index, if they are positive. Suppose at Date 0 you invest 100 in a p.p.n. with maturity T. Let r_S denote the return on an underlying S, typically an equity index. Your payoff at T is

$$\text{p.p.n. payoff} = \begin{cases} 100 + 100\rho r_S & \text{if } r_S > 0 \\ 100 & \text{if } r_S \leq 0. \end{cases}$$

The percentage ρ of the return r_S that the client receives is called the *participation rate* of the product.

The participation rate is a characteristic of a specific product that can be seen as a measure of attractiveness of capital guaranteed products. It summarizes some information about market characteristics, such as implied volatilities, interest rates, and dividends.

Throughout your solution, use notation above: refer to the participation rate as ρ, the (continuously compounded) risk-free rate as r, the time at inception as Date 0, and the maturity of the p.p.n. as Date T. Refer to the underlying as S, so $r_S = \frac{S_T - S_0}{S_0}$, and refer to the price and strike of a European call option written on S as c and K respectively.

1. Sketch the payoff of the p.p.n. above. Where is the kink in the payoff? What is the slope for $r_S > 0$?

2. Show graphically that the p.p.n. has the same payoff as the following:

   a) $100 + 100\rho \max\{0, r_S\}$.

   b) $100 + \frac{100\rho}{S_0} \max\{0, S_T - S_0\}$.

3. Draw the payoff diagram of the following portfolios:

   a) One call (with strike K).

   b) One bond and one call.

   c) Two calls.

   d) One bond and two calls.

4. The p.p.n. can be replicated with bonds and call options. Describe the replicating portfolio, i.e. find how many bonds are in the portfolio, how many calls, and what is the strike on the calls?

   Hints: (i) To find the number of calls in the portfolio, match the slope of the p.p.n. for $r_S > 0$. (ii) To find the strike of the calls, make sure the kink in the calls' payoff coincides with the kind in the p.p.n.'s payoff.

5. Suppose the structurer uses the 100 he gets when he sells the p.p.n. to hedge. If he buys 100 bonds with maturity T at Date 0, he has $100 - 100e^{-rT}$ to buy calls. Thus, he can make a riskless profit whenever

$$\underbrace{100 + \frac{100\rho}{S_0}\max\{0, S_T - S_0\}}_{\text{payoff of p.p.n.}} < \underbrace{100 + \frac{100 - 100e^{-rT}}{c_0}\max\{0, S_T - S_0\}}_{\text{payoff of hedging portfolio}}.$$

   Explain the terms in the expression above in a word or two.

6. Now, use the expression in the previous question to find the highest value of the participation rate ρ for which the structurer can make a certain profit.

   Hint: Just solve for ρ in terms of c_0, S_0, r, and T.

7. Why can the structurer promise this certain return to the customer and still make a certain profit himself?

8. What is the customer be missing/forgetting when he buys the product thinking it is a great deal?

9. Suppose the index pays dividends. Do higher dividends benefit the structurer or the customer (all else equal)? Why?

II. State Prices and Replication

Consider a binomial model with two assets, S^1 and S^2. State u and state d are equally likely. S^1 costs 90 today and pays off 120 in state u and 80 in state d. S^2 costs 110 today and pays off 80 in state u and 120 in state d. These payoffs can be written in vector notation as

$$\begin{pmatrix} S^1_u \\ S^1_d \end{pmatrix} = \begin{pmatrix} 120 \\ 80 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} S^2_u \\ S^2_d \end{pmatrix} = \begin{pmatrix} 80 \\ 120 \end{pmatrix}.$$

1. Find the prices ϕ_u and ϕ_d of the Arrow securities, i.e. the securities with payoffs

$$\mathbb{1}_{\{u\}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbb{1}_{\{d\}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

2. Find the portfolio that replicates the bond. Write down the risk-free rate r.

3. Use your answers to the questions above to write down the risk-neutral probability $\mathbb{Q}$.

4. What are
$$\frac{\mathbb{Q}[u]}{\mathbb{P}[u]} \quad \text{and} \quad \frac{\mathbb{Q}[d]}{\mathbb{P}[d]}\,?$$
Given your answer, which state, u or d, do you think is a recession?

5. Price the derivative X that pays off $X_T = \max\{S^1, S^2\}$.

III. The Binomial Model without a Risk-free Asset

1. Consider a binomial model with two securities, a volatile stock S^1 and a relatively stable stock S^2. S^1 pays off 120 in state u and 80 in state d and costs 100 today; S^2 pays off 100 in state u and 80 in state d and costs 90 today.

   In your solution, form a replicating portfolio with quantity Δ_1 of S^1 and a quantity Δ_2 of S^2.

   a) Price a derivative that pays off the greater of S^1 and S^2, i.e. has the payoff $X_T = \max\{S^1_T, S^2_T\}$.

   b) Price a call on S^1 with strike 110.

   c) Price a put on S^2 with strike 100.

2. Consider a binomial model with two securities, a corporate bond B and a credit default swap CDS. B pays off one in state u and zero in state d and costs B_0 today. CDS pays off zero in state u and one in state d and costs CDS_0 today.

   a) Write down an expression for the risk-free rate for this tree.

   b) Consider a general derivative that pays off X_u in state u and X_d in state d. Write down a general expression for the price X_0 of the derivative today. Your answer should depend on only B_0, CDS_0, X_u, and X_d.

      Hint: Proceed exactly as in question 1a.: Construct a replicating portfolio with Δ_1 corporate bonds and Δ_2 credit default swaps that matches the payoff of X both in state u and in state d.

   c) Write down an expression for the risk-neutral probability $\mathbb{Q}$ for this tree. I.e. find a number $q \equiv \mathbb{Q}[u]$ (so $1-q = \mathbb{Q}[d]$) such that

$$Y_0 = \frac{qY_u + (1-q)Y_d}{1+r}$$

      for any derivative Y. (Note that r is the risk-free rate you solved for in 2a.) Your answer—the expression for q—should depend on only B_0 and CDS_0.

IV. Optionality in Insurance

Read the February *FT* Alphaville post "Meet the man who could own Aviva France."

1. Describe the insurance product that L'Abeille Vie offered "its richer clients" in 1987; explain briefly how it caused trouble for Aviva recently.

2. Relate the product to the forward-start option you discussed in Problem Set 1.

3. Is the George family better off if markets are volatile or stable? Explain briefly but carefully.